Practice workbook · Rail Fence cipher

Zig-zag transposition: writing text over n rails, reading off ciphertext, finding the key from a diagram, naming the method and explaining why it is weak.

5 question parts: 1 worked-example retry and 4 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T3-20

1. Question 3 · part c.i · 4 marks

2020 NOV · Paper 02

Worked-example retry · Attempt part c.i only.

3 Several encryption algorithms have been developed.

(a) State what is meant by the term 'encryption'.

(1)

(b) Give **one** reason why data may need to be encrypted.

(1)

(c) (i) Perform a Rail Fence cipher encryption of the text **COMPUTATIONAL** using a key of 4.

Show your working.

(4)

Encrypted text:

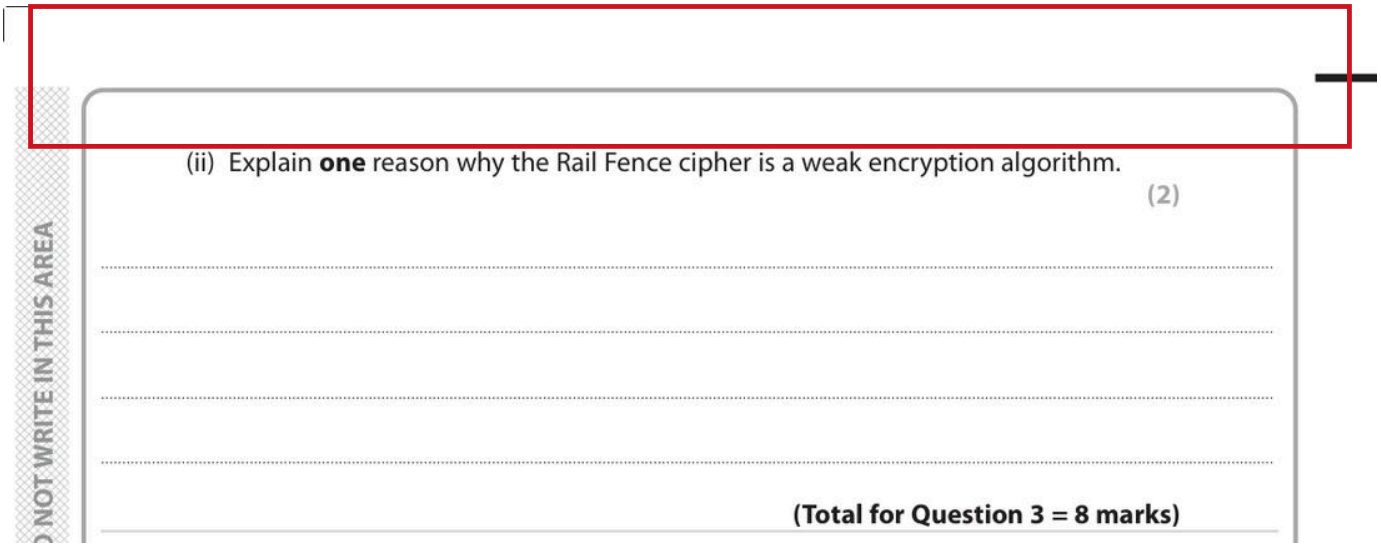
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P 6 1 8 8 5 A 0 6 1 6

1. Question 3 · part c.i · 4 marks

2020 NOV · Paper 02

Worked-example retry · Attempt part c.i only.



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(ii) Explain **one** reason why the Rail Fence cipher is a weak encryption algorithm.

(2)

(Total for Question 3 = 8 marks)

Working and answer

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2. Question 3 · part c.ii · 2 marks

2020 NOV · Paper 02

Further practice · Attempt part c.ii only.

3 Several encryption algorithms have been developed.

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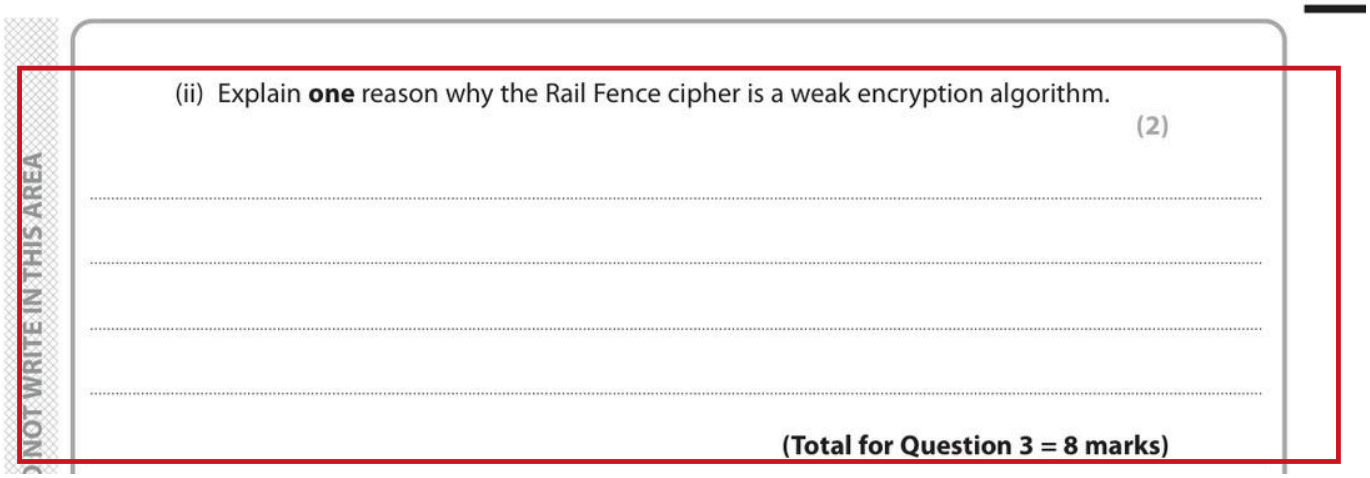
6

P 6 1 8 8 5 A 0 6 1 6

2. Question 3 · part c.ii · 2 marks

2020 NOV · Paper 02

Further practice · Attempt part c.ii only.


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Working and answer

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3. Question 4 · part c.i · 1 mark

2022 JUN · Paper 02

Further practice · Attempt part c.i only.

4 Several encryption algorithms have been developed.

(a) Identify what is meant by **encryption**.

(1)

- ☒ **A** Conversion of ciphertext into plaintext
- ☒ **B** Conversion of plaintext into ciphertext
- ☒ **C** Conversion of code into data
- ☒ **D** Conversion of data into information

(b) **Figure 1** shows a Pigpen cipher grid.

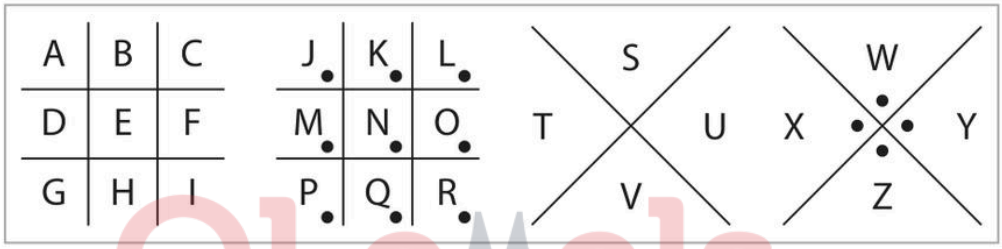


Figure 1

Complete the table by adding the symbol for each letter in the word **MAY**.

(3)

Letter	M	A	Y
Symbol			

(c) The message THE ENEMY IS NEAR is encoded using a different encryption algorithm.

Figure 2 shows the first stage of the encryption process.

T								I					
	H						Y		S				
		E				M				N			
			E		E						E		R
				N								A	

Figure 2

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3. Question 4 · part c.i · 1 mark

2022 JUN · Paper 02

Further practice · Attempt part c.i only.

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(i) Give the key used in the first stage of the encryption process.

(1)

(ii) Give the ciphertext that is produced by the encryption algorithm.

(1)

(iii) Give the name of the encryption algorithm that has been used to produce the ciphertext.

(1)

(Total for Question 4 = 7 marks)

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Working and answer

4. Question 4 · part c.ii · 1 mark

2022 JUN · Paper 02

Further practice · Attempt part c.ii only.

4 Several encryption algorithms have been developed.

(a) Identify what is meant by **encryption**.

(1)

- ☒ **A** Conversion of ciphertext into plaintext
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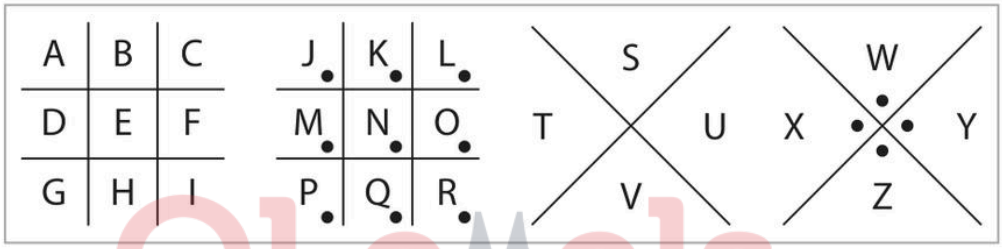


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		E				M				N			
			E		E						E		R
				N								A	

Figure 2

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P 7 2 4 0 6 A 0 8 1 6

4. Question 4 · part c.ii · 1 mark

2022 JUN · Paper 02

Further practice · Attempt part c.ii only.

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(i) Give the key used in the first stage of the encryption process.

(1)

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(Total for Question 4 = 7 marks)

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Working and answer

5. Question 4 · part c.iii · 1 mark

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Further practice · Attempt part c.iii only.

4 Several encryption algorithms have been developed.

(a) Identify what is meant by **encryption**.

(1)

- ☒ **A** Conversion of ciphertext into plaintext
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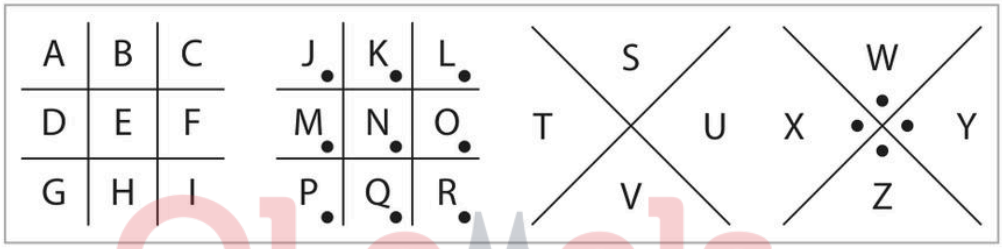


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T								I					
	H						Y		S				
		E				M				N			
			E		E						E		R
				N								A	

Figure 2

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DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

5. Question 4 · part c.iii · 1 mark

2022 JUN · Paper 02

Further practice · Attempt part c.iii only.

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(i) Give the key used in the first stage of the encryption process.

(1)

(ii) Give the ciphertext that is produced by the encryption algorithm.

(1)

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Working and answer